

CSC331: Data Structures - BMCC Fall 2025
Professor Byron
Program #3 specifications: Stack implementation
Due: 1159pm Fri 10-31-2025
No credit for late submission

Your task for this assignment is to implement a stack data structure in C++. This will be accomplished by utilizing a C++ user-defined class.

1. Implement a stack data structure using C++. The program will be interactive. Data transactions may be entered at the keyboard one by one or as a batch of transactions in a plain text file at the command line using redirection. The result of each transaction will be displayed on the console.

An example using redirection is the following: `$ prog3 < prog3.dat`

2. Each input transaction will contain an arithmetic expression in post-fix format. Assume that each operand and operation (+, -, *, /, %, ^) is a single character in each arithmetic expression without spacing. You may assume that each arithmetic expression is correctly formatted. However, your program should not attempt the following:
 - a. Divide by zero with / or % ... Instead, your program should display an error message and then begin evaluating a new input transaction.
 - b. Perform % with non-integer operands ... Instead, your program should display an error message and then begin evaluating a new input transaction.Note that the intermediate or final results of an expression may be negative or non-integer.

Sample input transactions and results:

- a. 23+ (result = 5)
 - b. 34*5/ (result = 2.4)
 - c. 832*6-/ (result = "error: division by zero")
 - d. 48*62*42^3+-% (result = 4)
 - e. 92/73/5%4/+ (result = "error: non-integer operand for %")
3. An input transaction containing "end-of-file" indicates there are no more transactions to be processed. Implement a stack to evaluate each expression and display the result. Use the C++ user-defined class named "stack331_f25.h" to implement stack functions. Use "include" in your program to retrieve the header file. No changes are permitted in the header file.
 4. The program will be run at the command prompt by navigating to the directory containing the executable version of the program after the program is compiled. The

program should display a prompt requesting input, such as “Please enter an expression in post-fix notation:”.

5. Your C++ program file should be named `csc331_section_prog3_lastname.cpp`. Your program should contain comments starting on line 1 of the program containing the following information:
 - a. course ID and section
 - b. your full name
 - c. the program file name
 - d. the program assignment number and due date
 - e. the program purpose

You are encouraged to add additional comments throughout the program that you feel might be helpful to the reader of your source code.

6. Submit your C++ program (cpp file) as an unzipped attachment to an email message to `kbyron@bmcc.cuny.edu` using a subject in this form: “`csc331_section_prog3_lastname`”. A link to a file located on a remote server will not be accepted. If you submit your assignment at least one week early, it will be graded prior to the submission deadline. If you wish, you may then re-submit your complete assignment prior to the due date for a potentially higher score.
7. Do your own work. Students submitting copies of the same program will receive grades of zero for the assignment. Do not share your code until you have received your final grade for the course at the end of the semester.
8. Grading rubric

Partial or extra credit will be awarded as follows:

0%: compiling errors

60%: significant logic with no compiling errors

90%: 5 of 6 operations working, but only one transaction processed

100%: 5 of 6 operations working, and terminated by “end-of-file” transaction

110%: 6 of 6 operations working, and terminated by “end-of-file” transaction